MILITARY SPECIFICATION

BEARING, SPIN

(SIDEWINDER)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the spin bearing critical item, referred to herein as the "spin bearing".

1.2 Classification. Spin bearings are of the following types:

Type I: Cartridge spin bearing.

Type II: Duplex spin-bearing pair.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A

FSC 3110

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SPECIFICATIONS

FEDERAL

QQ-S-763	Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting.
TT-T-656	Tricresyl Phosphate.

MILITARY

MIL-B-197	Bearings, Antifriction; Associated Parts and Subassemblies; Preparation for Delivery of.
MIL-B-1083	Balls, Bearing, Ferrous and Non Ferrous (for use in Bearings and Valves).
MIL-I-6868	Inspection Process, Magnetic Particle.
MIL-H-6875	Heat Treatment of Steel, Process for.
MIL-B-81793	Bearing, Ball, Precision, for Instruments and Rotating Components.
MIL-G-85127	Guidance and Control Sections, WGU-4A/B and WGU-4B/B Guided Missile (Sidewinder).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-206	Friction Torque Testing for Instrument Ball Bearings, Test Requirements, Parts 1 and 2.

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MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

SA492551	Sidewinder IC Vibration Test Fixture.
SA493245	Spin Bearing Preload Fixture.
SA2524273	Slow Down and Noise Test Fixture
1569600	Protector, Dome.
2412395	Sheet, Technical Frequency Specifications.
639AS3100	Seeker Section, Infrared, Guided Missile.
639AS3143	Bearing, Spin.
639AS3166	Retainer, Ball.
639AS3169	Gimbal Assembly.
639AS3180	Gyro Assembly.
639AS3195	Grease, Bearing.
639AS3962	Seeker Section, Infrared, Guided Missile.
639AS6502	Gyro Assembly.
639AS6522	Bearing, Spin.
639AS6523	Retainer, Ball.

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PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-874	Clean Room Conditions.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION (AFBMA)

AFBMA Standards	Annular Bearings Engineers Committee of the Anti-Friction Bearing Manufacturers Association (Twelve Sections).

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers Association, 1101 Connecticut Avenue, N.W., Suite 700, Washington D.C. 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	Surface Texture.
ANSI Y14.5	Dimensioning and Tolerancing.

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E45	Determining the Inclusion Content of Steel, Practice for.
ASTM E92	Vickers Hardness of Metallic Materials, Test Method for.

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ASTM E384	Microhardness of Materials, Test Method for.
ASTM DS56	Unified Number System (UNS) for Metals and Alloys.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS

Cataloging Handbook H4	Federal Supply Code for Manufacturers, United States and Canada.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Item description. The spin bearing is an inner race duplex preloaded instrument precision annular ball bearing. The spin bearing is of the following two types:

Type I. Cartridge spin bearing. The cartridge spin bearing consists of the following:

a. Double raceway grooves in flanged single outer bearing ring.

b. Two inner bearing rings with a single raceway groove in each ring.

c. Two ball cages in each cartridge bearing.

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d. Twelve balls in each raceway groove for a total of twenty-four balls in each cartridge bearing.

e. Open inseparable assembly.

Type II. Duplex spin-bearing pair. The duplex spin bearing pair consists of the following:

a. One plain (unflanged), single raceway groove, outer bearing ring.

b. One flanged, single raceway groove, outer bearing ring.

c. Two inner bearing rings with a single raceway groove in each ring.

d. Two ball cages in each duplex spin bearing.

e. Twelve balls in each individual bearing for a total of twenty-four balls in each duplex spin bearing.

f. Individual bearing assemblies are open and inseparable.

3.2 Characteristics.

3.2.1 Performance. The spin bearing shall meet the performance verification characteristics specified herein.

3.2.1.1 Low speed torque (Type I or Type II).

3.2 1.1.1 Nonpreloaded spin bearing. The individual inner rings or individual bearings, when operated as single bearings in conjunction with their respective outer ring raceway groove and lubricated with oil only in accordance with 3.3.2.2.3.2, shall display the following torque characteristics (see 6.4).

a. Average running torque: $\leq$3,000 milligram-millimeter (mg-mm).

b. Average hash width: $\leq$3,000 mg-mm.

c. Maximum hash (bearing contamination): $\leq$6,900 mg-mm.

d. Geometric wander: $\leq$1,600 mg-mm.

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e. Brinell damage (metal damage): No repeatable negative spikes 800 mg-mm greater than the average hash.

f. A nonrecurring positive excursion due to retainer hang-up is acceptable provided the duration is less than one-third revolution.

3.2.1.1.2 Relubrication for low speed dynamometer (LSD). In cases where the grease has been removed from the spin bearing and the sole purpose is to perform an LSD test to determine its condition after environmental testing, each spin bearing shall be lubricated either by impregnation in accordance with 3.3.2.2.3.2 or by adding two drops of the base oil of grease (see Drawing 639AS3195) prior to testing.

3.2.1.2 Running torque slowdown. The running torque of the spin bearing, lubricated in accordance with 3.2.1.1.1, shall cause the undriven rotor speed to decay from 7,200 to 3,600 revolutions per minute (rpm) in not less than 25 seconds.

3.2.2 Physical characteristics.

3.2.2.1 Balls. The balls shall have a nominal diameter of 0.062500 inch and shall be in accordance with the ball quality requirements of MIL-B-81793, Grade 3.

3.2.2.1.1 Ball complement. The cartridge and duplex bearing shall each have a complement of 24 balls. All balls for each individual bearing within the pair shall be taken from the same unit container.

3.2.2.2 Ball cage. The same ball cage (see Drawing 639AS3166 or 639AS6523) shall be used for both raceways in the cartridge or duplex bearing. The ball cage shall be inner-land piloted when used in the spin bearing in accordance with Drawing 639AS3143 and outer-land piloted when used in the spin bearing in accordance with Drawing 639AS6522 (see Figure 1).

3.2.2.2.1 Ball cage clearance (total running clearance).

3.2.2.2.1.1 Inner-land piloted (see Drawing 639AS3143).

a. Ball cage to inner raceway land: 0.004 to 0.008 inch.

b. Ball cage to outer raceway land: 0.015 inch nominal.

c. Ball cage to balls: 0.0020 to 0.0060 inch.

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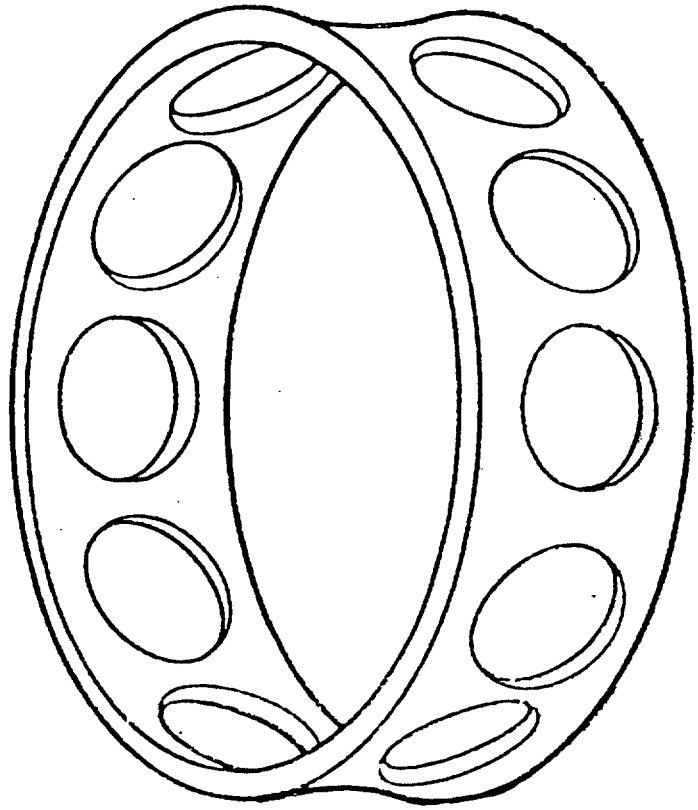


FIGURE 1. Ball cage. (See Drawing 639AS3166 or 639AS6523).

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3.2.2.2.1.2 Outer-land piloted (see Drawing 639AS6522).

a. Ball cage to outer raceway land: 0.004 to 0.008 inch.

b. Ball cage to inner raceway land: 0.015 inch nominal.

c. Ball cage to balls: 0.0020 to 0.0060 inch.

3.2.2.3 Rings. The rings shall be in accordance with Drawing 639AS3143 or 639AS6522 and AFBMA Standards (Section 12), ABEC-9P.

3.2.2.3.1 Wall thickness. The wall thickness at the thinnest cross-section between the raceway and the adjacent inside diameter for inner rings and the adjacent outside diameter for outer rings shall be not less than 0.011 inch.

3.2.2.4 Raceway radius. The cross-sectional (cross-race) radius of the raceways (race curvature) shall be 56 ±1 percent (0.035 inch nominal) of the nominal ball diameter (0.062500 inch). The departure from a true circle (DFTC) of the cross-race curvature within the working portion or contact area of the ball raceway shall be not greater than 0.000020 inch.

a. Contact angle for plain bearing shall be 15 to 30 degrees.

b. Contact angle for flanged bearing shall be 5 to 25 degrees.

3.2.2.4.1 Raceway groove runout. Raceway groove runout shall be not greater than 0.000050 inch in accordance with ABEC-9P tolerances as follows:

a. Outer rings with respect to 0.4062 inch diameter.

b. Inner rings with respect to 0.2344 inch diameter.

c. Race-to-face for all rings with respect to clamping surface -C- (see Drawing 639AS3143 or 639AS6522).

d. The cartridge flange face shall be used in lieu of -C- for cartridge bearing.

e. Outer raceway parallelism in the outer ring of the cartridge bearing shall be not greater than 0.000050 inch between the two raceways.

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3.2.2.4.2 Raceway roundness. The individual circumferential roundness of the inner ring bearing raceway and outer ring bearing raceway, respectively, shall fall within a 0.000025 inch wide annular tolerance zone in accordance with ANSI Y14.5; that is, a fifty-millionths bearing.

3.2.2.5 Individual bearings.

3.2.2.5.1 Inseparable bearings. Individual bearings or inner rings shall be inseparable and shall not separate when a backload of 2.00 ±0.05 pounds is applied through the inner raceway, against the ball dam, at an angle of 0 degree ±15 minutes to the spin reference axis.

3.2.2.5.2 Relative position of bearing components. When the individual bearing is measured under a preload of not less than 1.25 pounds and not greater than 2.00 pounds, the relative positions of the rings and ball cage shall be in accordance with Drawing 639AS3143 or 639AS6522 (see 3.2.2.2). The inner ring mating face plane of a mated pair of spin bearings is not required to be coincident with the outer ring mating face plane.

3.2.2.5.3 Radial play. Radial play shall be as follows:

a. Flanged bearing shall have 0.0006 to 0.0008 inch radial play.

b. Plain bearing shall have 0.0010 to 0.0012 inch radial play.

3.2.2.6 Spin bearings.

3.2.2.6.1 Coding. Both bearings of the spin bearing, Type I or II, shall meet the requirements of 3.2.2.5.2 and shall have the same code designation. Coding shall be in accordance with Drawing 639AS3143 or 639AS6522 and AFBMA Standards (Section 12). The size code shall be determined by the maximum outside diameter (OD) and the minimum inside diameter (ID). Type I cartridge is treated as a single code outer with matching code inner rings.

3.2.2.6.2 Preloaded pair. The individual bearings or inner rings shall be selected so that the spin bearing shall have a relative DB preload of 1.25 to 2.00 pounds (see Figure 2 and 3.2.2.6.2.1.1) when installed as shown on Figure 2 and in Table I. Preload selection for a given set of bearing components shall be compatible with the running torque requirements of 3.2.1.2.

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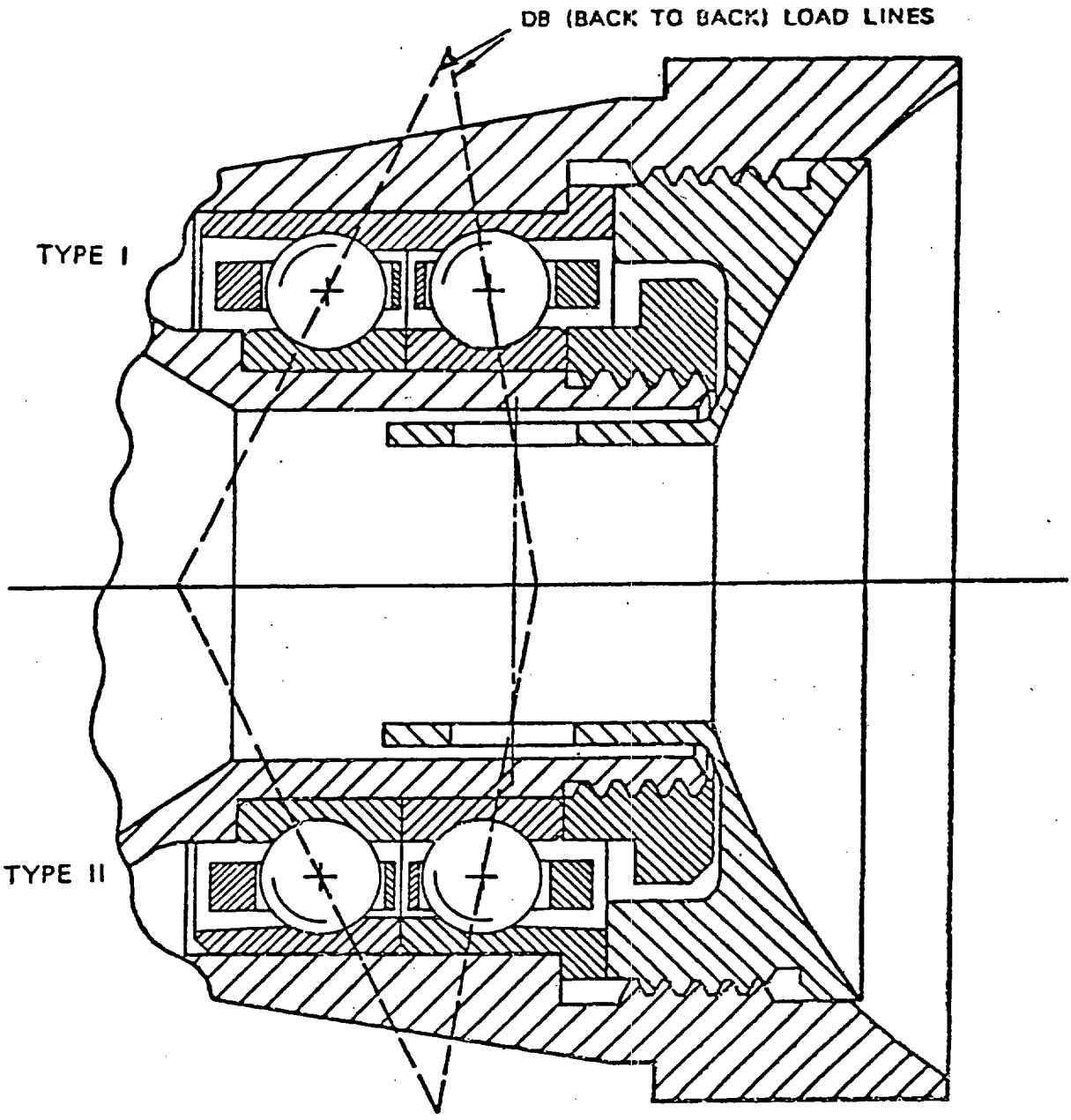


FIGURE 2. DB preload diagram.

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TABLE I. Mass properties, fit, and clamping forces.

Gyro or dummy rotor mass (gm)	Gyro or dummy rotor polar moment of inertia ($gm\text{-}cm^2$)	Maximum unbalance (micro-ounce-inch)	Bearing shaft clearance (micro-inch)1/	Bearing housing clearance (micro-inch)1/	Inner ring clamping torque (inch-ounces) (based on 0.2344-64 thread)
107 ±2	348 ±3	150	30-80	50-150	30-34

1/ Bearing clearances are determined by using the minimum ID of the hole and the maximum OD of its mating part.

3.2.2.6.2.1 Preload curve plot. The preload curve plot of load (abscissa) versus deflection (ordinate) shall clearly indicate the relative preload present in the bearing pair (see Figure 3), regardless of ring orientation. Curve tolerance regardless of preload shall be as shown on Figure 3.

3.2.2.6.2.1.1 True preload. True preload (see Figure 3) exists when both inner races of the bearing assembly are clamped together in accordance with Table I.

3.2.3 Environmental.

3.2.3.1 Vibration. Upon completion of the vibration test the spin bearing shall meet all the requirements of 3.2.1. The average running torque and average hash shall not change greater than 1,000 mg-mm from the original torque value measured during acceptance testing.

3.2.3.2 Shock. Upon the completion of the shock test (both operating and nonoperating) the spin bearing shall meet all the requirements of 3.2.1. The average running torque and average hash shall not change greater than 1,000 mg-mm from the original torque value measured during acceptance testing.

3.2.3.3 Operating life. The spin bearing shall demonstrate not less than 650-hour life at a 90 percent confidence level. Following the operating life test, the spin bearing shall meet all the requirements of 3.2.1. The average running torque and average hash shall not change greater than 1,000 mg-mm from the original torque value measured during acceptance testing.

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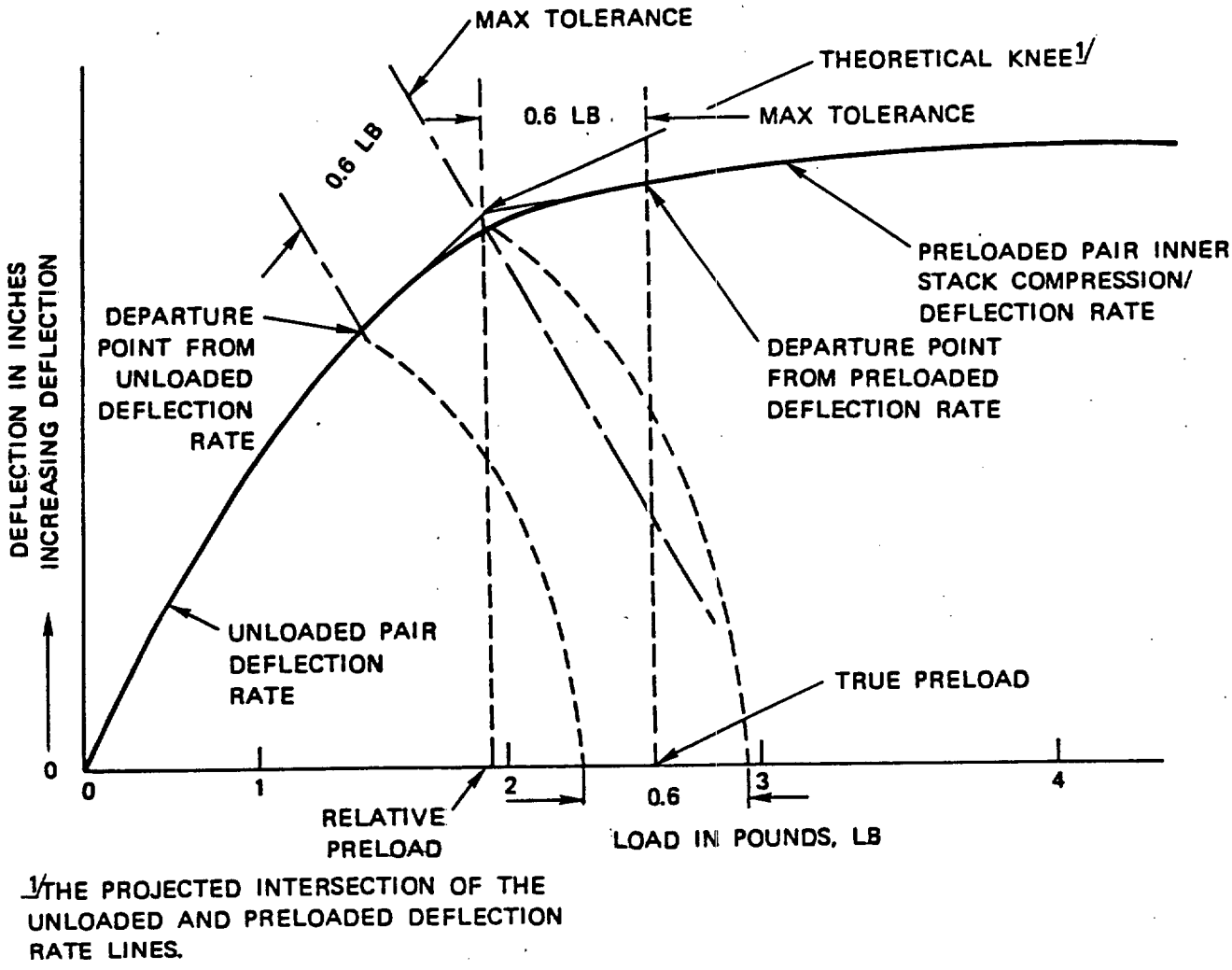


FIGURE 3. Preload curve tolerance.
(Related to Drawing SA493245.)

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3.3 Design and construction.

3.3.1 Production drawings. The spin bearing shall be manufactured, fabricated, and assembled in accordance with Drawing 639AS3143 or 639AS6522 and as specified herein. Details of the parts not specified by this specification and Drawing 639AS3143 or 639AS6522 shall be optional with the contractor.

3.3.2 Standards of manufacture.

3.3.2.1 Materials.

3.3.2.1.1 New materials. New materials shall be used in the manufacture of component parts and fabrication of bearings furnished in accordance with this specification.

3.3.2.1.2 Reconditioned component parts. Spin bearings purchased as new items in accordance with this specification shall not contain component parts that have previously been installed in any equipment and operated.

3.3.2.1.3 Porosity control. Fabrication of the ball cage shall employ manufacturing techniques that do not degrade the inherent porosity of the material.

3.3.2.1.4 Metals. The inner and outer rings shall be made from consumable-electrode, vacuum-melted stainless steel. All steel used in the rings and balls shall be in accordance with QQ-S-763, Class 440 C, Condition A (UNS S44004 in accordance with ASTM DS-56).

3.3.2.1.5 Phenolic. The ball cage shall be made of phenolic tubing in accordance with Drawing 639AS3166 or 639AS6523.

3.3.2.2 Processes.

3.3.2.2.1 Heat treatment. Heat treatment shall be in accordance with MIL-H-6875.

3.3.2.2.2 Passivation. The rings and balls shall be passivated, after the last material removal operation, in accordance with MIL-B-81793.

3.3.2.2.2.1 Surface treatment. After passivation and prior to assembly, the rings and balls shall be treated with Tricresyl Phosphate (TCP) $((CH_3C_6H_4O)_3PO)$ in accordance with TT-T-656 to ensure uniform distribution of lubricant on the balls and within the raceways (no oil beading).

3.3.2.2.3 Lubrication.

3.3.2.2.3.1 Grease lubricant. The grease lubricant shall be in accordance with Drawing 639AS3195.

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3.3.2.2.3.1.1 Contamination. The grease lubricant contamination shall be in accordance with MIL-B-81793.

3.3.2.2.3.2 Nonmetallic retainer impregnation. Prior to grease application, the bearing rings and cage shall be thoroughly cleaned, vacuum dried, and vacuum impregnated in accordance with MIL-B-81793 using the same base lubricating oil (Trifluoropropylmethyldimethylpolysiloxane) used in accordance with Drawing 639AS3195, except that excess oil shall be removed by centrifuging bearing at 175 ±25 gravity units (g) with spin axis parallel to direction of force for 1.0 ±0.2 minute.

3.3.2.2.3.3 Grease application. Each individual assembled bearing of the bearing pair shall be packed with 8 to 10 milligrams (mg) of grease. The grease shall be in accordance with Drawing 639AS3195. Grease shall be used in the neat (straight, undiluted) condition. It shall be applied during low-speed rotation using a suitable syringe with a number 30 needle to ensure uniform distribution. Each raceway complement of the cartridge bearing (Type I) shall be lubricated as an individual bearing, that is, 8 to 10 mg per raceway complement for a total of 16 to 20 mg per cartridge evenly distributed.

CAUTION

Grease shall not be diluted with any solvent or spreading agent. Excess oil applied for torque and slowdown tests shall be removed by centrifuging bearings for not less than 1 minute at 175 ±25 g.

3.3.2.2.4 Solvent purity. Only high purity reagent grade solvents with a residue not greater than five parts per million (ppm) shall be used in all processes involving finished bearings or finished components.

3.3.2.3 Parts.

3.3.2.3.1 Hardness. When heat treated in accordance with 3.3.2.2.1, the rings shall be through-hardened to a minimum hardness of Rockwell C58. The balls shall be through-hardened to a minimum hardness of Rockwell C59.

3.3.2.3.1.1 Fracture grain size. Fracture grain size shall be 7.5 or finer.

3.3.2.3.1.2 Cracks and fractures. Ring and ball integrity shall be in accordance with MIL-B-81793.

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3.3.2.3.1.3 Dimensional stability. Dimensional stability shall be in accordance with MIL-B-81793.

3.3.2.4 Surface finish of rings and balls. Careful examination shall reveal no evidence of decarburization, rust, tool marks on the mounting surfaces, scratches across the races, residual embedded abrasives, or Brinell marks on the load-bearing surfaces (see 3.3.3).

3.3.2.5 Surface finish of ball cages. Careful examination shall reveal no visible defects such as cracks, feather edges, ragged edges, dangling fibers longer than 0.005 inch, or loose particles of phenolic bonding material (see 3.3.3).

3.3.2.6 Surface roughness of rings. The surface roughness of the rings shall be not greater than the following:

a. Raceway surfaces: 3 microinches arithmetical average (AA). (Cross-race lapping may be employed to achieve this finish.)

b. Mounting surfaces (outside and inside diameters plus clamping and mounting faces (see Figure 2): 8 microinches AA.

c. Cage pilot lands: 8 microinches AA.

d. Nonfunctional internal surfaces: 32 microinches AA.

e. Corners, chamfers, fillets, and flange OD: 63 microinches AA.

3.3.2.7 Visual and mechanical. The visual requirements of the spin-bearing characteristics shall be in accordance with MIL-B-81793, except as specified herein.

3.3.2.8 Reliability. Spin bearing reliability shall be in accordance with 3.2.3.3.

3.3.2.9 Interchangeability. The spin bearings shall be interchangeable within any given size, code, or dash number (see Drawing 639AS3143 or 639AS6522). Individual rings and balls are not required to be interchangeable from one spin bearing to another.

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3.3.2.10 Product marking. The spin-bearing package shall be permanently and legibly identified and marked in accordance with MIL-STD-130 and Drawing 639AS3143 or 639AS6522. Marking shall include, but not be limited to, the following:

a. NAVAIR Drawing 639AS3143 or 639AS6522.

b. NAVAIR Code Identification 30003.

c. Manufacturer's name and symbol or code identification from Federal Cataloging Handbook H4.

d. A unique serial number.

e. Dash number and size code; for example, Drawing 639AS3143-1, A-A or Drawing 639AS6522-1, A-A.

3.3.2.10.1 Ring identification. Outer faces of inner rings or flanged bearing shall be marked with the manufacturer's identification by electro-etch, acid-etch, or soft stamp in accordance with the marking for identification of MIL-B-81793, except as specified herein.

3.3.2.10.2 Post marking. Identification marking shall not degrade functional surfaces nor impair functional integrity.

3.3.3 Material cleanliness. Material cleanliness shall be in accordance with MIL-B-81793, except inclusion levels shall be as indicated below:

	Type	J-K rating
a.	Sulfide:	Thin series 1.5; heavy series 1.0
b.	Alumina:	Thin series 1.0; heavy series 1.0
c.	Silica and/or silicate:	Thin series 1.0; heavy series 1.0
d.	Globular oxides:	Thin series 1.5; heavy series 1.0

3.3.3.1 Component cleanliness. All tools and fixtures used to handle the bearings and all bearing components shall be clean and free from any chemical or foreign contamination.

3.3.4 Special working conditions. The facilities used in the manufacture, testing, and inspection of the spin bearing shall be in accordance with AD-874.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.1.2.1 and 6.2.3).

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3.5 Workmanship. Workmanship, finishing, cleaning, and lubrication of spin bearing pairs shall be performed in a manner that will assure compliance with all requirements of this specification and all applicable referenced documents. The contractor shall be in accordance with AD-874 and MIL-B-81793 to assure production of high quality bearings.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 Classification of inspections. The inspection and testing of the bearings shall be in accordance with the following classifications:

a. First article inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

4.1.1.2 Test equipment. Test equipment shall be of sufficient accuracy, quality, and quantity to ensure that the performance for the required tests and measurements are to the accuracy specified herein. All equipment shall be calibrated and maintained in accordance with a system meeting the requirements of MIL-STD-45662.

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4.1.2 First article inspections.

4.1.2.1 First article inspection sequence. The first article sample (see 3.4) shall be subjected to the following inspection sequence:

a. All 12 spin bearings shall be subjected to all of the acceptance tests shown in Table II, Group I.

b. Six spin bearings shall be selected after completion of the Table II, Group I tests and subjected to all the product-verification tests shown in Table II, Group II.

c. Four of the spin bearings used for the Table II, Group II tests shall be further subjected to the environmental tests shown in Table II, Group III.

d. The remaining two spin bearings used for product-verification tests shall be disassembled and the components used for the Table II, Group IV tests. Each of the component tests shown in Table II, Group IV shall be performed on the applicable components of both spin bearings.

e. A total of six spin bearings from the remaining first article pairs shall be subjected to the operating life test of Table II, Group V.

f. Any spin bearing which fails any test shall be rejected.

In the event of a failure during any stage of the first article inspection, failure analysis shall be used to determine if the failure should be attributed to poor manufacturing, design, or to some unrelated cause. Additional tests shall be performed after failure correction to verify that acceptable bearings are being produced and the failure mode eliminated. The contracting activity shall be satisfied that the bearing manufacturer can and does meet the intent and requirements of this specification.

4.2 Quality conformance inspections. The production lot sample shall be subjected to the inspections of Table II, 4.2.1, and 4.2.2.

4.2.1 Inspection lot acceptance criteria. Each spin bearing shall be subjected to all the tests shown in Table II, Subgroup 1, in accordance with 4.2.4.1. Any bearing and spin bearing which fails any Subgroup 1 acceptance tests shall be rejected. Each spin bearing of Subgroup 2 acceptance test sample selected in accordance with 4.2.4.1 shall be subjected to the running torque test of Table II, Subgroup 2. If any failure occurs, the failed spin bearings shall be rejected and each spin bearing of the lot represented by the sample shall be subjected to the running torque test of Table II, Subgroup 2.

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TABLE II. First article and quality conformance inspections.

Inspection group	Inspection	Requirement paragraph	Method paragraph
Group I			
Acceptance tests			
Subgroup 1	Lubrication	3.3.2.2.3	4.3.4.4.3
	Workmanship	3.3.1, 3.3.2.10, 3.5	4.3.5
	Coding	3.2.2.6.1	4.3.2.1
	Preload	3.2.2.6.2	4.3.2.2
	Low speed torque	3.2.1.1	4.3.1.1.1
Subgroup 2	Running torque	3.2.1.2	4.3.1.2
Group II			
Product-verification tests	Inseparable bearings	3.2.2.5.1	4.3.2
	Relative position of bearing components	3.2.2.5.2	4.3.2
	Radial play	3.2.2.5.3	4.3.2
Group III			
Environmental tests	Vibration	3.2.3.1	4.3.3.1
	Shock	3.2.3.2	4.3.3.2
Group IV			
Component-parts tests	Rings, dimensions, and tolerances	3.2.2.3, 3.2.2.3.1	4.3.2
	Raceway radius and runout including DFTC for both cross-race and circumferential roundness	3.2.2.4, 3.2.2.4.1, 3.2.2.4.2	4.3.2
	Ball, dimensions, and tolerances	3.2.2.1	4.3.2
	Ball complements	3.2.2.1.1	4.3.2
	Ball cage	3.2.2.2	4.3.2
	Ball cage porosity	3.3.2.1.3, 3.3.2.2.3.2	4.3.4.1.1
	Surface finish, rings and balls	3.3.2.4	4.3.4.3.2

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TABLE II. First article and quality conformance inspections. - Continued

Inspection group	Inspection	Requirement paragraph	Method paragraph
Group IV - Continued			
	Surface finish, cages	3.3.2.5	4.3.4.3.3
	Surface roughness, rings	3.3.2.6	4.3.4.3.4
	Passivation	3.3.2.2.2	4.3.4.2.1
	Dimensional stability	3.3.2.3.1.3	4.3.2
	Hardness	3.3.2.3.1	4.3.4.3.1
	Materials, rings and balls	3.3.2.1, 3.3.2.1.1, 3.3.2.1.2, 3.3.2.1.4	4.3.4.1.2
	Materials, ball cage	3.3.2.1.5	4.3.4.1.3
	Wettability of rings and balls	3.3.2.2.2.1	4.3.4.2.2.1
	Decarburization	3.3.2.4	4.3.4.3.2.1
	Fracture grain size	3.3.2.3.1.1	4.3.4.3.1.1
	Cracks and fractures	3.3.2.3.1.2	4.3.4.3.1.2
	Material cleanliness	3.3.1	4.3.5
Group V			
Service life	Operating life	3.2.3.3	4.3.3.3

4.2.2 Product verification tests. Each spin bearing of the product verification sample, selected in accordance with 4.2.4.2, shall be subjected to all the tests of Table II, Group II. If any spin bearing fails any Table II, Group II tests, the sample shall be rejected and a new sample shall be drawn and tested. If any failures occur during the testing of the second sample, reject the failed bearing and subject the entire lot represented by the two samples to the test or tests in which the failures occurred.

4.2.2.1 Component tests.

4.2.2.1.1 Prior to assembly. Component parts selected in accordance with 4.2.4.3 shall be subjected to the applicable tests shown in Table II, Group IV. Acceptance or rejection of the component lot shall be in accordance with 4.2.4.3.

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4.2.2.1.2 Test after assembly. Two or more spin bearing pairs from each qualification sample shall be dismantled and subjected to the tests shown in Table II, Group IV. Bearings previously used in product-verification tests and forwarded to the testing activity in accordance with 4.2.4.2 shall also be subjected to the Table II, Group IV tests, at the discretion of the contracting activity. Should any component fail to meet the requirements of the Table II, Group IV tests, the contracting activity shall have the right to retest the bearings or withdraw the qualification of the product, or to take such other remedial action as shall be deemed appropriate.

4.2.3 Lot. A lot shall consist of all spin bearings made of the same materials group and in the same production run which are offered for delivery at the same time. A lot of components shall consist of all like components from the same machine run or production process being submitted for inspection at the same time.

4.2.4 Sampling procedures.

4.2.4.1 Sampling for acceptance tests. Sampling for acceptance tests of Subgroup 1 shall be on a 100-percent basis. Subject each spin bearing to the tests of Table II, Subgroup 1. Sampling for acceptance test of Subgroup 2 shall be in accordance with MIL-STD-105, Special Inspection Level S-3, Single Sampling Plan, Acceptable Quality Level (AQL) Level 4.0. See 4.2.1 for defective rate.

4.2.4.2 Sampling for product-verification tests. The product-verification sample shall consist of 4 percent of all spin bearings, randomly selected from bearings which have been tested and have passed all the acceptance tests. Samples shall be subjected to the tests of 4.2.2. After completion of the product-verification tests, the sample of spin bearings shall be forwarded to the design activity.

4.2.4.3 Sampling for component-parts tests. Sampling of component parts prior to assembly shall be in accordance with MIL-STD-105, Inspection Level II, Double Sampling Plan. The sample lot shall be accepted or rejected in accordance with AQLs shown in Tables III and IV for the designated category and classification of critical, major, and minor defects in accordance with MIL-B-81793. Sampling of parts of assembled bearings for product verification or qualification shall be as specified in the applicable test requirements.

TABLE III. Categories of defects and AQLs by tolerance.

Tolerance	Defect category	AQL (percent)
ABEC-9P[1/]	Critical	0.65
	Major	1.00
	Minor	2.50

[1/] As defined in AFBMA standards.

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TABLE IV. Classification of defects.

Category	Code	Description	Requirement
Critical	1	Cracks or fractures in any component	3.3.2.3.1.2
	2	Missing parts - rings, balls, cage components	3.3.1
	3	Incorrect material	3.3.2.1.4
Major	101	Outer ring OD	3.2.2.3
	102	Outer ring OD out-of-round	3.2.2.3
	103	Outer ring OD taper	3.2.2.3
	104	Outer ring radial runout	3.2.2.3
	105	Outer ring cross-race curvature out of specification	3.2.2.4.2
	106	Outer ring OD runout with reference face (flanged or unflanged)	3.2.2.3
	107	Outer ring raceway roundness out of specification	3.2.2.4.2
	108	Outer ring OD/flange face undercut	3.3.2.6.e
	109	Outer ring raceway runout to reference side (both flanged and unflanged OD)	3.3.2.6.e
	110	Outer ring raceway surface finish exceeds specification	3.3.2.4
	111	Inner ring bore out-of-round	3.2.2.4.2
	112	Inner ring radial runout	3.2.2.3
	114	Inner ring bore taper	3.2.2.3
	115	Inner ring bore runout with reference side	3.2.2.3 3.2.2.4.1b
	116	Inner ring raceway runout to reference side	3.2.2.4.1 3.2.2.4.2
	117	Inner ring raceway surface roughness exceeds specification	3.2.2.4.2
	118	Inner ring raceway roundness out of specification	3.2.2.4.2
	119	Inner ring cross-race curvature out of specification	3.2.2.3
	120	Inner ring corner radii	3.2.2.3
	121	Lubricant - not as specified	3.3.2.2.3
	122	Lubricant quantity or lubrication not as required	3.3.2.2.3
	123	Cross-corner (0.1720) dimension out of specification	Drawing 639AS3143 or 639AS6522

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TABLE IV. Classification of defects. - Continued.

Category	Code	Description	Requirement
Major cont.	124	Component hardness - not within required limits	3.3.2.3.1
	125	Radial play out of specification	3.2.2.5.3
	126	Cage delamination of nonmetallic separator ball pockets	3.3.2.7
	127	Ball diameter variation not within specified limits	3.2.2.1
	128	Balls do not meet visual requirements	3.3.2.7
	129	Ball surface finish exceeds specification	3.2.2.1
	130	Bearing parts do not meet requirements of specification	3.3.2.2.1
Minor	201	Outer ring width variation	3.2.2.3
	202	Outer ring flange width	3.2.2.3
	203	Outer ring flange OD	3.2.2.3
	204	Outer ring corner radii	3.2.2.3
	205	Outer ring surface roughness on mounting surface exceeds specification	3.3.2.4
	206	Outer ring land surface roughness exceeds specification	3.3.2.4
	207	Inner ring width variation	3.2.2.3
	208	Inner ring surface finish on mounting surface exceeds specification	3.3.2.4
	209	Inner ring land surface finish exceeds specification	3.3.2.4
	210	Raceways do not meet visual requirements	3.3.2.7
	211	Cages do not meet visual requirements (other than Major 126)	3.3.2.7
	212	Cage clearance missing	3.2.2.2.1
	213	Mounting surfaces do not meet visual requirements	3.3.2.7
	214	Bearing marking, bearing identification, package marking, or mounting instructions for duplex pairs are incomplete or missing	3.3.2.10 3.3.2.10.2.1
	215	Bearing packaging or packing not as specified (for coding only refer to MIL-B-197 for inspection details)	MIL-B-197
	216	Bore and/or OD size coding is incomplete or missing	3.2.2.6.1

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4.3 Test methods.

4.3.1 Performance.

4.3.1.1 Low speed torque. Lubrication for the torque trace test shall be in accordance with 3.3.2.2.3.2. (Data shall be recorded on a chart with a scale sensitivity of 400 mg-mm or less for each division.) The bearing rotation speed and the chart speed combination shall be one of the following conditions:

Condition	A	B	C
Bearing speed (rpm)	1/2	1	2
Chart speed (mm/mm)	50-60	120-150	120-150

4.3.1.1.1 Type I. The individual inner rings when operated as single bearings shall be tested in accordance with MIL-STD-206 at a 400-gram load. The torque trace shall meet the requirements of 3.2.1.1.1.

4.3.1.1.2 Type II. The individual bearings (plain or flanged) of the duplex bearing shall be tested in accordance with MIL-STD-206 at a 400-gram load. The torque trace shall meet the requirements of 3.2.1.1.1.

4.3.1.2 Running torque slowdown. The spin bearing shall be lubricated in accordance with 3.3.2.2.3.2, except the centrifuge level shall be 1,000 ±25 g's for 5 minutes, and installed in a slowdown and noise fixture (see Drawing SA2524273 or equal slowdown test fixture approved by the contracting activity) using a dummy rotor having properties, fit, and clamping forces shown in Table I, except the bearing shaft and housing clearances shall be 100-300 microinches. The running torque of the bearing shall meet the requirements of 3.2.1.2.

4.3.2 Physical. The bearings shall be subjected to the appropriate tests of either AFBMA Standards or MIL-B-81793 and shall meet the physical requirements of 3.2.2.

4 3.2.1 Coding. The bearing shall be measured in accordance with gaging practices specified in AFBMA Standards (Section 12) for instrument ball bearings. The spin bearing shall meet the requirements of 3.2.2.6.1.

4.3.2.2 Preloaded pair. The preload shall be measured on the bearing preload fixture (see Drawing SA493245 or equal preload test fixture approved by the testing activity) with the bearing lightly lubricated as specified in 3.3.2.2.3.2. The preload shall meet the requirements of 3.2.2.6.2.

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4.3.2.2.1 Preload plot. Figure 3 is an illustration of a hypothetical preload plot consisting of two AA straight lines connected by a curve that starts and stops with 0.6 pound on either side of the tangent intersection point (preload point). The bearings shall be rejected when the curved portion of the plot starts and stops outside the limits shown on Figure 3 regardless of preload (see 3.2.2.6.2.1.1).

4.3.3 Environmental. The environmental tests of Table II, Group III, shall be performed on bearings selected from the first article sample and on other bearings forwarded to the testing activity in accordance with 4.2.4.2. Failure of the production sample to meet the requirements specified herein will be cause for further investigation and possible disqualification of the vendor.

4.3.3.1 Vibration. The spin bearing shall be assembled to a gimbal, (see Drawing 639AS3169) which is then installed in a gyro (see Drawing 639AS3180 or Drawing 639AS6502) in accordance with the fit, balance, clamping force, and other parameters of Table I, and installed in a seeker assembly (see Drawing 639AS3100 or Drawing 639AS3962 for the vibration test). The seeker assembly shall be mounted on a fixture approved by the cognizant design activity (see 6.2.1 and 6.3). The gyro shall be rotated in accordance with frequency F_1 of Drawing 2412395 throughout the vibration test, which will be conducted at an ambient temperature of 21 ±5 degrees Celsius (°C). The seeker assembly shall be subjected to random vibration of 10 minutes duration along each of three mutually perpendicular axes, with inputs as shown on Figure 4. The spin bearing shall meet the requirements of 3.2.3.1 at the conclusion of these tests.

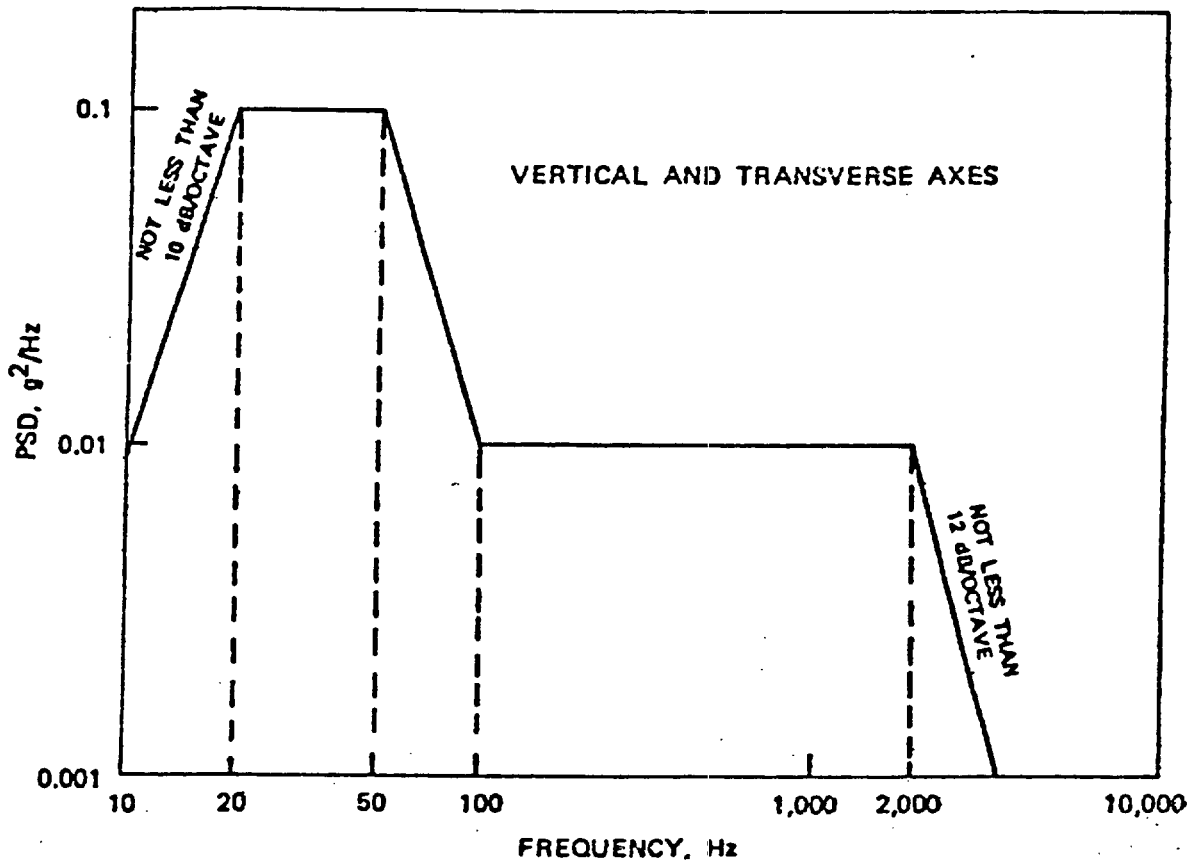
4.3.3.2 Shock. Shock tests shall be divided into two phases, nonoperating and operating.

4.3.3.2.1 Mounting. The spin bearing shall be installed in accordance with 4.3.3.1 with adaptability to shock tests along each of the three mutually perpendicular axes. The acceleration of all shocks shall be measured on the aft end of the seeker housing.

4.3.3.2.2 Procedure for shock test (nonoperating). A standard dome protector (see Drawing 1569600), shall be firmly secured to the front end of the seeker head in order to provide magnetic caging on the nonoperating gyro. For this test, the gyro shall possess the magnetic characteristics of a production gyro. Subject the bearing, installed in accordance with 4.3.3.1, to 12 shocks, two in each direction, along each axis (four per axis) as shown in Table V and on Figure 5.

4.3.3.2.3 Procedure for shock test (operating). Repeat the test of 4.3.3.2.2 with the gyro or dummy gyro operating at the spin frequency in accordance with 4.3.3.1, except the dome protector shall be removed for this test. After completion of the test, the bearing shall meet the requirements of 3.2.3.2.

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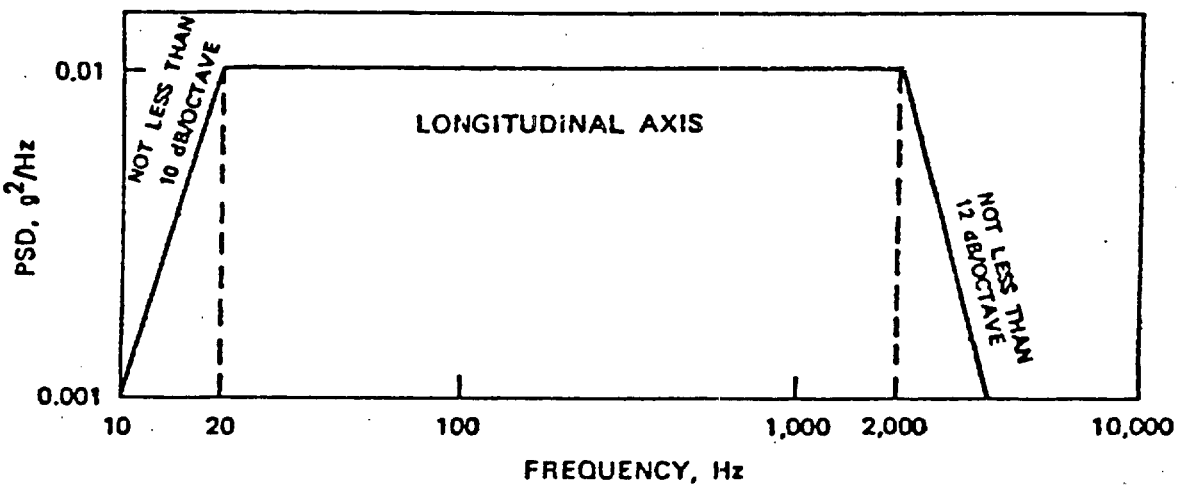


FIGURE 4. Vibration spectrum.

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TABLE V. Shock test.

Direction	Acceleration peak (g) A	Duration of acceleration (milliseconds)	Number of operating shocks	Number of non-operating shocks
Along spin axis	47.5 ±2.5	21.0 ±3.0	4	4
Along outer gimbal axis	47.5 ±2.5	21.0 ±3.0	4	4
Along inner gimbal axis	47.5 ±2.5	21.0 ±3.0	4	4

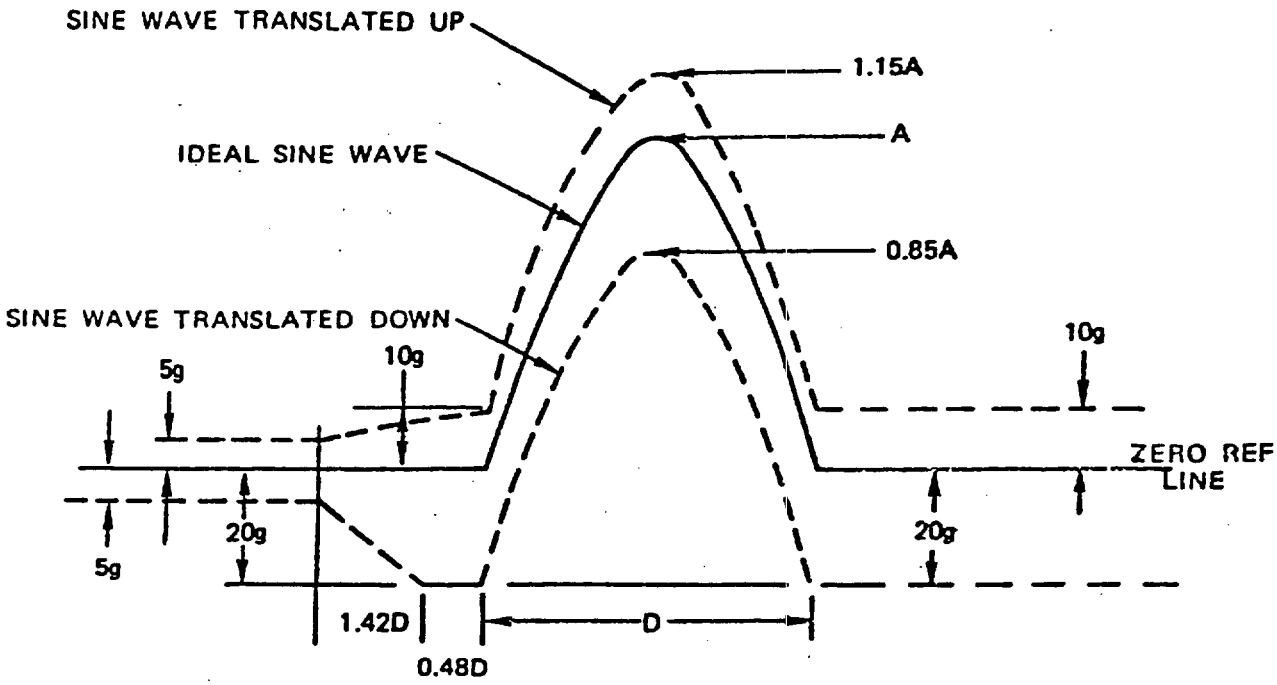
4.3.3.3 Operating life test. The spin bearing shall be mounted in a seeker assembly (see Drawing 639AS3100 or Drawing 639AS3962), which shall be subsequently mounted in a simulated GCS, which is mounted in a vibration test fixture fabricated in accordance with Drawing SA492551. The geometric center of the test fixture shall be located on the vibration exciter center line and shall be subjected to the rotation tests of 4.3.3.3.1, the vibration tests of 4.3.3.3.2, and the temperature tests of 4.3.3.3.3. At the conclusion of the operating life test, each spin bearing shall meet the requirements of 3.2.3.3.

4.3.3.3.1 Operating life test rotation. The operating life test shall consist of 650 accumulative hours of rotation for each spin bearing, which shall be accrued during 39 24-hour cycles as shown on Figure 6 (see 3.2.3.3).

4.3.3.3.2 Operating life test vibration. Vibration shall be applied to the rotating test unit in accordance with Figure 6. The vibration input shall be as shown on Figure 6, Note 4.

NOTE: Prior to commencing vibration testing, cross-axis acceleration shall be measured at the accelerometer locations in accordance with Drawing SA492551 with an input to the fixture in accordance with 4.3.3.3.2. Cross-axis acceleration measurements shall be conducted with a simulated GCS installed in fixture (see Drawing SA492551). The cross-axis response shall be not greater than 50 percent of the input vibration at any nonresonant mode (1:2).

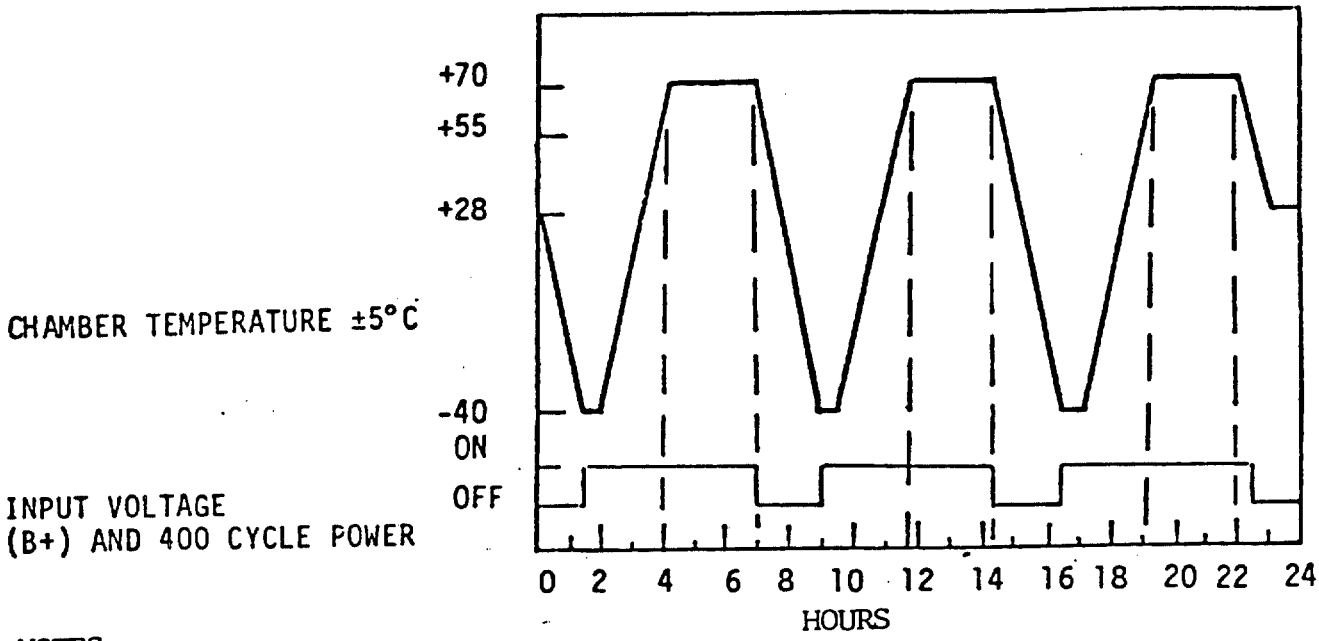
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NOTE:
THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE ACCELERATION AMPLITUDE OF THE IDEAL HALF SINE PULSE IS A AND ITS DURATION IS D. THE MEASURED ACCELERATION PULSE SHALL BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES AND THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS $V_i \pm 0.1V_i$ WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS $2AD/\pi$. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4D BEFORE THE PULSE TO 0.1D AFTER THE PULSE

FIGURE 5. Shock test.

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NOTES:

1. A THERMAL SURVEY MAY BE CONDUCTED IN ACCORDANCE WITH MIL-STD-781 ON ONE UNIT AND NEED NOT BE REPEATED AGAIN. TEMPERATURE STABILIZATION FOR THE UPPER TEMPERATURE (+70°C) ONLY SHALL BE IN ACCORDANCE WITH THE RESULTS OF THE THERMAL SURVEY AND SHALL APPLY TO EACH EXCURSION TO THE PROFILE UPPER TEMPERATURE.

2. INPUT VOLTAGES WILL BE IN ACCORDANCE WITH MIL-G-85127(AS).

3. INPUT VIBRATION LEVEL MONITORED ON VIBRATION FIXTURE.

4. THE VIBRATION PROFILE FOR THE TEMPERATURE CYCLE SHALL BE APPLIED DURING EACH HOUR OF TEST IN WHICH THE GCS IS ENERGIZED. THE PROFILE SHALL BE AS FOLLOWS:

VIBRATION DURATION (MINUTES)	INPUT PSD LEVEL (g^2/Hz)	FREQUENCY (Hz)
50	0.005	20 TO 2000
3	0.015	20 TO 2000
1	0.05	20 TO 100

FIGURE 6. Temperature cycling.

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4.3.3.3.3 Operating life test temperature. The seeker assembly shall be mounted in a test chamber in which the 24-hour temperature cycle shown on Figure 6 shall be repeated throughout the life test.

4.3.3.4 Failure analysis. Should any bearing undergoing life testing develop excessive noise, torque, or other suspicious symptoms (see 4.3.3.3), the test may be interrupted until the bearings have been examined. If a decision is made to discontinue testing of such bearings, a complete failure analysis shall be performed. Corrective action shall be subject to approval by the contracting activity for all failures.

4.3.4 Materials, processes and parts.

4.3.4.1 Materials.

4.3.4.1.1 Oil retention. Sampling for inspection of oil retention shall be in accordance with MIL-STD-105, Special Instruction Level S-3, Single Sampling Plan, AQL 4.0 percent defective. Oil retention and bleed rate shall be tested in accordance with MIL-B-81793, appendix paragraphs as follows:

a. Retainer preparation.

b. Vacuum inpregnation process.

c. Oil retention and bleed rate test.

d. Oil retention test.

e. Bleed rate test.

4.3.4.1.2 Metals. The materials used in the rings and balls shall be tested in accordance with MIL-B-1083, except the magnetic particle inspection shall not be performed. The amount and type of microscopic inclusions shall be determined using the ASTM inclusion chart, Plate III, Designation E-45. The material shall meet the requirements of 3.3.2.1.5 and 3.3.3. Certified copies of test reports will be accepted as evidence of compliance with these requirements.

4.3.4.1.3 Phenolic. The phenolic material used in the cage shall meet the requirements of 3.3.2.1.5. Certified copies of test reports and other data from the original source of supply may be accepted as evidence of compliance with this requirement.

4.3.4.2 Processes.

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4.3.4.2.1 Passivation. The rings and balls shall be tested in accordance with MIL-B-81793 and shall meet the requirements of 3.3.2.2.2.

4.3.4.2.2 Surface treatment.

4.3.4.2.2.1 Wettability. TCP treated rings and balls shall be individually tested for lubricant wettability. Sampling for inspection of wettability shall be in accordance with MIL-STD-105, Special Inspection Level S-3, Single Sampling Plan, AQL Level 4.0. Rings and balls shall be dipped in an 85:1 solution of freon and oil (see 3.3.2.2.3.2) and baked at 74 ±5°C for 15 to 20 minutes to remove freon solvent. Examine individual raceways and balls for oil beading or retraction. The rings and balls shall meet the requirements of 3.3.2.2.2.1.

4.3.4.3 Parts.

4.3.4.3.1 Hardness. The test procedures and conversion tables of ASTM E92 or ASTM E384, as applicable, shall be used in testing the rings and balls for hardness. The Rockwell C hardness shall be obtained by converting from the Diamond Pyramid Hardness (DPH) as measured on the flat surface of a properly prepared metallographical specimen, using a 500-gm load or by an acceptable equal method. Not less than five readings shall be taken on each ring, and the average value converted to the Rockwell C hardness. Not less than five balls from the same bearing shall be measured, one reading being taken near the center of the flat place on each ball, and the second reading to be taken near the edge of the flat. The hardness of the parts shall meet the requirements of 3.3.2.3.1.

4.3.4.3.1.1 Fracture grain size. Fracture grain size test shall be in accordance with MIL-B-1083 for composition 2 balls. The balls shall meet the requirements of 3.3.2.3.1.1.

4.3.4.3.1.2 Cracks and fractures. Test for cracks and fractures using magnetic particle inspection shall be in accordance with MIL-I-6868. The following test conditions shall apply:

a. Circular and longitudinal magnetization to uncover all possible flaws.

b. Continuous method of particle application.

c. Wet, visible particles.

Indications of surface or subsurface cracks, fractures, or seams shall reject the lot in accordance with 3.3.2.3.1.2.

4.3.4.3.2 Surface roughness of rings and balls. When inspected for roughness-height rating in accordance with ANSI B46.1, using a 0.001-inch cutoff width and a 0.0001-inch radius diamond stylus, the rings and balls shall meet the requirements of 3.2.2.1 and 3.3.2.6, respectively.

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4.3.4.3.2.1 Decarburization. A sample of balls and rings shall be selected in accordance with MIL-STD-105, Single Sampling Plan, Inspection Level S-4, AQL 0.4 percent defective. Transverse sections through the center of each sample shall be polished, microetched, and examined at a magnification of 100 power (X). If any of the test specimens exhibits surface decarburization, the lot shall be rejected.

4.3.4.3.3 Surface finish of ball cages. The clean, dry ball cages shall be examined under a 30X magnification microscope and the ball cages shall meet the surface finish requirements of 3.3.2.5.

4.3.4.3.4 Surface finish of rings. The clean dry ring shall be examined under a 30X magnification microscope and the rings shall meet the requirements of 3.3.2.6.

4.3.4.3.5 Abrasive examination. Inspect raceways and balls with a 100X microscope illuminated with polarized light. Abrasive particles will show up as bright spots in the dark field. Components shall be free of residual embedded abrasives in accordance with 3.3.2.4. Alternative inspection techniques shall be subject to approval by the cognizant design activity.

4.3.4.4 Lubrication.

4.3.4.4.1 Contamination. Lubricant contamination shall be determined in accordance with MIL-B-81793 and shall meet the requirements of 3.3.2.2.3.1.1.

4.3.4.4.2 Nonmetallic retainer impregnation. The nonmetallic retainer shall be tested in accordance with MIL-B-81793. The contractor shall certify for the contracting activity that the requirements of 3.3.2.2.3.3 have been met.

4.3.4.4.3 Grease application. The contractor shall certify for the contracting activity that the requirements of 3.3.2.2.3.3 have been met.

4.3.5 Visual and mechanical. The bearing shall be examined for conformance to 3.3.1, 3.3.2.10 and 3.5.

4.3.6 Preservation, packaging and marking. Visual examination shall be made prior to shipment to determine conformance with Section 5.

5. PACKAGING.

5.1 Preservation - packaging.

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5.1.1 Level A. The individual bearings of the spin bearing pair and data card (see 5.3.1) shall be separately packaged in clean polyethylene and heat sealed in accordance with MIL-B-197, Method 1A-8. The packages shall be sealed for shipment in a foil-lined kraft bag.

5.1.1.1 Alternate method. Pairs of spin bearings shall be packaged in plastic vials in accordance with MIL-B-197, Method 1A-13, with suitable buffer material to prevent movement of the spin bearing pairs in the vial. The vial shall be marked with a pressure-sensitive label (see 5.3.1). Each vial shall be sealed in plastic celon which encloses the pressure-sensitive label.

5.2 Packing.

5.2.1 Level C. Spin bearings, packaged as specified in 5.3.1, shall be packed in accordance with MIL-B-197, domestic shipment for immediate use.

5.3 Marking.

5.3.1 Packaging. Each spin bearing pair shall be accompanied by a data card or label. The data card or label shall contain the following (complete with data):

Manufacturer's name _____

Manufacturer's part number _____

Drawing number _____

Preload _____

Bore/OD codes and/or dash number _____

Lubrication _____

Date _____

System use _____(specify AIM-9M)

Any other information requested by the contract or purchase order (see 6.2.1).

5.3.2 Shipping containers. In addition to any special markings required by the contract or purchase order (see 6.2.1), shipping containers shall be marked in accordance with MIL-STD-129, except there is no NSN for this bearing.

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6. NOTES

6.1 Intended use. The bearings specified herein are intended for use as a DB-mounted, preloaded, rotor spin bearing in a gyro optical system used on Guidance and Control Section, WGU-4A/B or WGU-4B/B.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grease or oil lubrication (see 3.3.2.2.3 and 4.3.4.4.3).

c. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

d. Sampling, if other than as specified in 4.2.4.

e. Approval of test fixture (see 4.3.3.1).

f. Packaging information, if other than as specified in 5.3.1.

g. Special markings on shipping containers (see 5.3.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID no.	Option
4.3.3.4	Failure report	DI-R-21598	---
4.3.4.1.2, 4.3.4.1.3	Test reports	DI-T-2072	---
4.3.4.4.2, 4.3.4.4.3	Certification Data/ Report	UDI-A-23264	---

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(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5012.12L., AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of 12 spin bearings. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Mounting for shock and vibration. Mounting of the bearing is in accordance with MIL-STD-2081 and MIL-STD-2082.

6.4 Definitions.

6.4.1 Torque test. Torque test definitions that are not included in MIL-STD-206.

6.4.1.1 Average hash width. The average bandwidth of the bearing torque observed during test cycle.

6.4.1.2 Bearing contamination. The torque value of the largest single positive spike peak-to-peak observed during test cycle.

6.4.1.3 Metal damage (Brinell). The torque value of the largest negative single peak-to-peak observed during test cycle.

6.4.1.4 Geometric wander. The variation in average torque during a test cycle, and is the difference between the maximum and the minimum instantaneous average torque during a test cycle.

6.4.2 Bearing and/or spin bearing. These terms are used interchangeably.

6.5 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.6 Supersession data. This specification supersedes the following:

a. MIL-B-85107A, dated 26 February 1985.

b. MIL-B-85107A, ECP M9078R1 (NOR M9078.1A), dated 10 January 1984.

c. MIL-B-85107A, ECP F1134R2 (NOR F1134.1B), dated 26 April 1984.

d. MIL-B-85107A, ECP F3274R1 (NOR F3274.1), dated 26 November 1984.

e. MIL-B-85107A, ECP G0014R1 (NOR G0014.1A), dated 18 September 1985.

f. MIL-B-85107A, ECP (NOR).

6.7 Subject term (keyword) listing.

AIM-9M
Bearing, spin
Components, missile
Guidance, missile
Sidewinder
WGU-4A/B
WGU-4B/B

Preparing activity:
Navy (AS)
(Project 3110-N005)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85107B(AS)	BEARING, SPIN (SIDEWINDER)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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